

TAKAROA, French Polynesia

WMO: 919430

Lat: 14.454S

Lon: 145.025W

Elev: 3

StdP: 101.29

Time Zone: -10.00 (W10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	23.8	24.3	18.2	13.1	26.0	19.1	13.9	26.3	13.2	25.7	12.2	26.1	4.7	70	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	3.6	31.8	26.3	31.3	26.0	31.0	25.7	27.2	30.4	26.7	30.1	26.4	29.9	4.6	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.7	29.6	25.7	21.0	29.3	25.3	20.5	29.1	85.7	30.2	83.8	30.2	82.3	30.1	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
11.3	10.3	9.5	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			22.5	32.2	0.5	1.0	22.1	32.9	21.9	33.5	21.6	34.0	21.2	34.7
			19.4	27.2	0.7	0.9	18.9	27.8	18.4	28.3	18.0	28.8	17.5	29.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.9	28.3	28.5	28.8	28.8	28.2	27.5	27.0	26.7	27.0	27.4	27.9	28.1
	DBStd	1.10	0.91	0.98	0.95	0.83	0.78	0.75	0.73	0.81	0.77	0.84	0.82	0.97
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6517	568	518	584	563	564	526	526	517	511	541	538	562
	CDD18.3	3475	309	284	326	313	306	276	267	259	261	282	288	304
	CDH23.3	38069	3614	3391	3958	3750	3486	2886	2546	2352	2476	2899	3214	3498
	CDH26.7	10882	1208	1214	1515	1386	1071	639	388	335	418	646	929	1133
(14) Wind	WSAvg	5.3	5.3	5.3	5.4	4.9	4.9	5.6	5.8	5.6	5.8	5.4	5.0	4.8
(15) Precipitation	PrecAvg	1495	201	176	134	115	92	95	73	68	79	122	183	199
	PrecMax	2260	601	606	394	452	463	319	299	170	219	279	612	630
	PrecMin	697	55	43	16	6	7	3	23	15	11	22	32	17
	PrecStd	310	116	115	79	98	76	67	50	38	53	61	115	130
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.2	32.2	32.6	32.2	31.3	30.8	29.7	29.8	30.2	30.7	31.2	31.7
		MCWB	26.3	26.7	27.3	26.3	25.7	25.2	24.6	24.3	24.5	25.2	25.8	26.3
	2%	DB	31.1	31.4	31.7	31.5	30.7	30.0	29.3	29.2	29.6	30.2	30.7	31.1
		MCWB	25.6	26.1	26.4	26.0	25.5	25.0	24.3	24.1	24.3	24.9	25.5	25.8
	5%	DB	30.6	30.9	31.1	31.0	30.3	29.5	28.9	28.8	29.2	29.7	30.2	30.6
		MCWB	25.4	25.8	26.1	25.8	25.4	24.8	24.2	24.0	24.2	24.7	25.3	25.5
	10%	DB	30.2	30.4	30.6	30.5	29.9	29.1	28.5	28.4	28.7	29.3	29.8	30.2
		MCWB	25.3	25.6	25.9	25.7	25.3	24.6	24.0	23.8	24.0	24.6	25.1	25.3
	0.4%	WB	27.6	27.7	27.9	27.2	26.7	26.1	25.6	25.2	25.5	26.1	26.6	27.1
		MCDB	30.3	31.2	31.7	30.4	30.2	29.4	28.6	28.7	29.0	29.7	30.2	30.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	26.7	26.9	27.1	26.7	26.2	25.7	25.1	24.9	25.0	25.6	26.1	26.6
		MCDB	29.6	30.0	30.5	30.1	29.6	28.9	28.2	28.2	28.5	29.1	29.6	30.1
	5%	WB	26.2	26.4	26.6	26.3	25.9	25.3	24.8	24.6	24.7	25.2	25.8	26.1
		MCDB	29.6	29.9	30.1	29.8	29.3	28.5	27.8	27.8	28.1	28.6	29.2	29.6
	10%	WB	25.7	26.0	26.2	26.0	25.7	25.1	24.6	24.3	24.4	25.0	25.5	25.8
		MCDB	29.4	29.7	29.8	29.7	29.0	28.2	27.6	27.5	27.7	28.3	29.0	29.2
(35) Mean Daily Temperature Range	5% DB	MDBR	3.8	3.8	3.6	3.6	3.4	3.2	3.1	3.3	3.5	3.6	3.7	3.9
		MCDBR	4.2	4.1	3.9	3.9	3.8	3.5	3.5	3.7	3.9	4.0	4.1	4.3
		MCWBR	1.8	1.8	1.7	1.7	1.7	1.7	1.7	1.7	1.8	1.8	1.8	1.9
	5% WB	MCDBR	3.9	3.9	3.8	3.8	3.5	3.3	3.2	3.4	3.7	3.8	3.9	4.1
		MCWBR	1.9	2.0	1.8	1.8	1.7	1.7	1.8	1.7	1.8	1.8	1.9	2.0
(40) Clear-Sky Solar Irradiance	taub	0.382	0.381	0.382	0.381	0.373	0.369	0.368	0.368	0.376	0.374	0.378	0.383	(40)
	taud	2.616	2.625	2.616	2.630	2.649	2.639	2.623	2.617	2.592	2.618	2.628	2.615	(41)
	Ebn at Noon	961	954	934	902	877	863	873	901	925	950	959	959	(42)
	Edh at Noon	103	101	99	93	86	84	87	92	100	101	101	103	(43)
(44) All-Sky Solar Radiation	RadAvg	6.51	6.42	6.26	5.69	4.93	4.62	4.86	5.52	6.24	6.62	6.66	6.55	(44)
	RadStd	0.40	0.57	0.32	0.38	0.37	0.21	0.20	0.20	0.29	0.34	0.35	0.53	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page